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Mental Workload Assessment from ECG and PPG Signal Using Machine Learning Techniques

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Abstract— Chronic stress is closely associated with the onset of mental health disorders and cardiovascular conditions, underscoring the need for effective stress monitoring to support overall well-being. Despite this, many individuals frequently ignore stress indicators, which can lead to significant health complications. Recent neuroscience research identifies mental stress as a major health crisis of the 21st century, yet standardized methods for its assessment are lacking. Although cortisol levels in saliva have traditionally served as an indicator, advancements in wearable technologies have drawn attention to physiological markers, such as heart rate variability (HRV), for stress evaluation. This study proposes a supervised machine learning approach to detect mental stress by analyzing HRV. Data were gathered from 22 participants who completed the N-back task, a cognitive workload test, while their electrocardiogram (ECG) and photoplethysmography (PPG) signals were recorded. HRV features were extracted and labeled using NASA-TLX scores to quantify mental workload. Various machine learning models with cross-validation were applied, yielding high accuracy rates of 96% and 88% with the random forest classifier, demonstrating a promising approach for effective stress detection.

Keywords— *Mental stress detection, Machine learning, N-back task, NASA-TLX, SVM, Logistic Regression*

I. INTRODUCTION

The degree of information processing capability or resources required by an operator to meet system demands and chores is referred to as mental workload (MW). A high mental workload suggests that more resources or information processing capability are required to execute a task or activity [1]. Human operators' information processing capacities can be better understood with the use of the MW evaluation. That is why MW evaluation is so important when avoiding task errors or working in an "overload" situation [2]. Researchers and medical professionals have been trying to establish methods for evaluating mental stress. Physiological signals, including electroencephalograms (EEGs), electrocardiograms (ECGs), galvanic skin responses (GSRs), photoplethysmography, and others, are commonly employed to quantify mental workload.

People's mental stress has increased a lot because of the significant changes in society in the past few years [3]. Although moderate mental stress can enhance attentional regulation and enhance learning and productivity, excessive

stress can have detrimental effects on both physical and psychological well-being. Fatigue is a consequence of long-term psychological strain. The body generates physiological signs or symptoms as a response. Disregarding these signals increases the susceptibility of the immunological and neural systems to potentially detrimental conditions, including cancer, cardiovascular disease, depression, and neurological disorders [4]. Moreover, a significant determinant of emotional instability or suicide is profound psychological distress. So, it's important to be aware of and deal with stress to improve our quality of life and lower our physical and mental health risk.

This study uses two signals (ECG and PPG) with machine learning algorithms like Random-forest, Decision-tree, Logistic Regression, Linear SVC, Gaussian naïve-bayes, KNN (K-Nearest Neighbor) to detect mental stress. EEG-based techniques provide precise evaluations of mental workload in intricate environments, whereas ECG[5] and PPG-based approaches yield moderate accuracy in less complex situations. Therefore, we anticipate that ECG and PPG-based methods will be the primary choice for continuous mental workload assessment. The N-back task is utilized as a means to quantify a component of working memory, frequently applied as an evaluation in the fields of psychology and cognitive neuroscience. In this test, a larger number of N indicates a greater mental burden. Here, three-class and four-class classification techniques have been proposed to detect mental stress.

II. METHODOLOGY

A. Dataset Acquisition:

The MAUS dataset compiles easily accessible physiological signals collected under various levels of mental demand, established using the N-back task to create distinct mental workload conditions. This dataset is especially useful for developing mental workload assessment systems, with a focus on PPG-based wearable devices. It includes signals from ECG, Fingertip-PPG, Wrist-PPG, and GSR, allowing users to directly compare Fingertip-PPG and Wrist-PPG data. In total, the dataset features 35-minute recordings from 22 participants (2 females) capturing these physiological signals [18]. It has twenty-two healthy students from the same institution, including two women. Participants were positioned in a room that maintained a consistent temperature, controlled lighting,

and little noise. They were sat at a table equipped with devices to record physiological signals, a computer monitors to show tasks, and a keyboard to respond to stimuli. As shown in Table I, two devices were used in this database to collect data from four channels at the same time. The Pro-comp Infinity enables accurate and precise real-time measurement of many channels with exceptional signal quality. The device provides an internal calibration process that is enabled by the user. This calibration process is considered the highest standard and guarantees the production of the best possible signal and for that reason we have used signal from this device in our study. The Pix-Art PPG [6][7] watch transmits the signal from a wearable device.

TABLE I. SUMMARY OF COLLECTED DATA

Subjects	Collected Signal	Frequency
22(20 males and 2 females)	ECG	256
	PPG	256
	Pix-art PPG	256
	GSR	100
<i>Data collection step</i>		
N-back task	0-back: low MW; 2,3-back: high MW	
Trial duration	7 min. (5 min. N-back + 2 min. Resting)	
No of trials	6 (0, 2, 3, 2, 3, 0 -back)	
Total duration	47 min. (7 min.*6 trials + 5 min. Resting)	

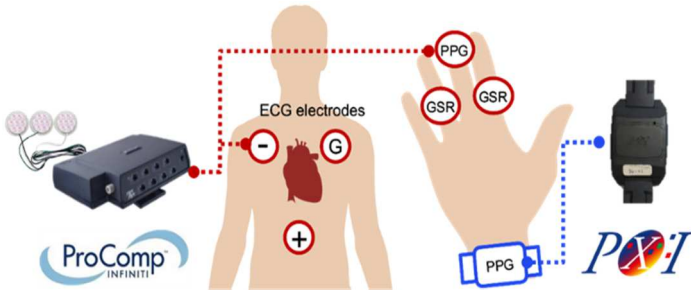


Fig. 1. Positioning of different sensors [18]

It provides the capability to explore data processing techniques for wearable devices. Figure 1 illustrates the precise locations where the electrodes are positioned to acquire peripheral physiological data.

B. Stress Detection Techniques :

1) N-back task:

At the outset of the experiment, participants engaged in a five-minute rest period and completed the Pittsburgh Sleep Quality Index (PSQI) questionnaire. This Sleep Quality Index is linked to cognitive workload, and the PSQI serves as a widely recognized tool for evaluating and quantifying sleep quality.[8][9] The n-back task is a frequently employed continuous performance assessment in the fields of psychology and cognitive neuroscience. It serves as an evaluation tool to evaluate a specific aspect of cognitive ability and working memory capacity. The task consists of six trials of the N-back task following a rest period [10]. In the N-back task, the participant needed to remember the most recent digits in a quickly presented sequence. The participant pressed the space bar to respond whenever the current stimulus

matched the one presented n steps earlier. The intricacy of the work increased proportionally with the value of N . Additional numerical values needed to be acquired, and the memorized sequence had to be constantly modified. Consequently, altering the value of N might trigger the activation of MW at different degrees. There will be three categories of tasks: 0-back, 2-back, and 3-back. The 2-back and 3-back tasks imposed a higher mental workload (MW) than the 0-back task, which was regarded to have a lower MW condition. The six trials were presented in a counterbalanced order as follows: 0-2-3-2-3-0[11]. Prior to the initial experiment, guidelines were provided specifying the type of work to be performed, which included 0-, 2-, or 3-back tasks. A random single-digit number was displayed every 3 seconds after the subject pressed the space bar to proceed. Each number was displayed for a duration of 2 seconds. During the experiment, a set of 25 items, representing 25% of the total, was used. In all, 100 numbers were displayed. The performance assessment involved recording both the replies and the response time. Following each exercise, participants took a two-minute break to individually evaluate the difficulty of the task.

2) Post-task NASA-TLX:

After each n-back activity, a subjective evaluation was conducted, as previously determined. This stage involved the implementation of the NASA Task Load Index (NASA-TLX) questionnaire [12]. It is a widely used subjective evaluation tool that measures the perceived difficulty of a work across many dimensions. The NASA-TLX questionnaire comprises six subscales designed to assess mental workload. These subscales are mental demand, physical demand, temporal demand, performance, effort, and frustration. The scoring for each subscale ranges from 0 to 100, with increments of five points. In order to determine the individual weighting of each subscale, the subject was required to compare and evaluate the six subscales in pairs, considering their evaluated relevance, after getting the six scores. A weight was assigned to each subscale depending on its frequency of selection. Each subscale score was multiplied by its assigned weight, and the weighted scores were then summed and divided by 15 to produce a total score, ranging from 0 to 100. The outcome represented MW's thorough subjective assessment of the effort.

C. Data Collection from Stress Techniques:

1) Analysis of user performance and subjective ratings:

Looking at user performance and subjective evaluations it was confirmed that individuals did sense various mental workload levels across different n-back circumstances. The sensors and their attachment locations are shown in Fig. 1.

2) Measurement of performance:

The six ratings were combined to provide a weighted NASA-TLX score, representing the total subjective evaluation burden. According to post-hoc analyses, MW during (2,3-back) activities was considerably greater than during (0-back) tasks, indicating that tasks with higher N demand more mental effort than those with lower N . Overall, when N increased, all six subscales expanded simultaneously[13].

III. WORKING PROCEDURE

The overall workflow for this paper is represented in Fig 2. The chronology was followed to process raw data, extract the necessary features, and classify.

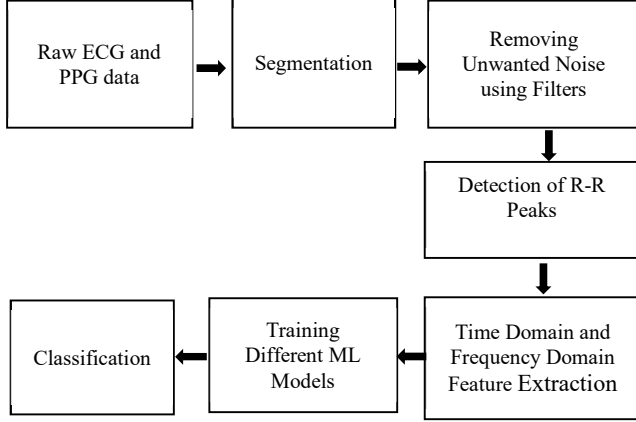


Fig. 2. Graphical demonstration of Working Procedure

A. Preprocessing:

The acquired ECG and PPG data require thorough preprocessing before feature extraction to ensure accuracy and reliability in subsequent analyses. Preprocessing primarily involves applying filters and smoothing techniques to the signals. For ECG signals, a bandpass filter is used, while PPG signals benefit from median filtering. Specifically, ECG signal preprocessing utilizes a Butterworth bandpass filter with cutoff frequencies of 0.5 Hz and 50 Hz, effectively capturing the essential frequency components while removing baseline wander and high-frequency noise. In contrast, PPG signals, which generally contain lower-frequency components, are filtered within a 0.1 Hz to 4 Hz range using a Butterworth bandpass filter to isolate the relevant information and reduce noise.[17] All preprocessing steps are conducted using MATLAB software, leveraging its specialized built-in functions that are particularly effective for digital signal processing tasks, including filtering and smoothing.

B. Detection of peak:

To extract heart rate variability (HRV) features, the initial step is detecting the R-R interval, accomplished through a peak detection method. Various techniques are available for this, including the Pan & Tompkins and Wavelet methods. In this study, the Pan & Tompkins algorithm was applied to detect ECG peaks, as shown in Fig. 3, using MATLAB's built-in functions. The R-R series, derived from both ECG and PPG signals, holds essential information for HRV analysis. For feature extraction, statistical measures such as mean, median, variance, standard deviation (STD), and the root mean square of successive differences (RMSSD) were calculated from the R-R intervals. Specifically, the PPG signal was first preprocessed using a median filter [12] before applying a peak detection algorithm. Accurate peak identification is critical to reliable HRV estimation. To enhance ECG signal quality, additional filtering was performed before peak detection. Following R-R series detection, several HRV features were extracted, as detailed in

Table III, with equations for select features provided below. Accurate R-R peak detection is crucial for extracting features.

The correct identification and precise location of peaks are essential for deriving meaningful features. An example of peak detection using MATLAB is demonstrated in Fig. 3.

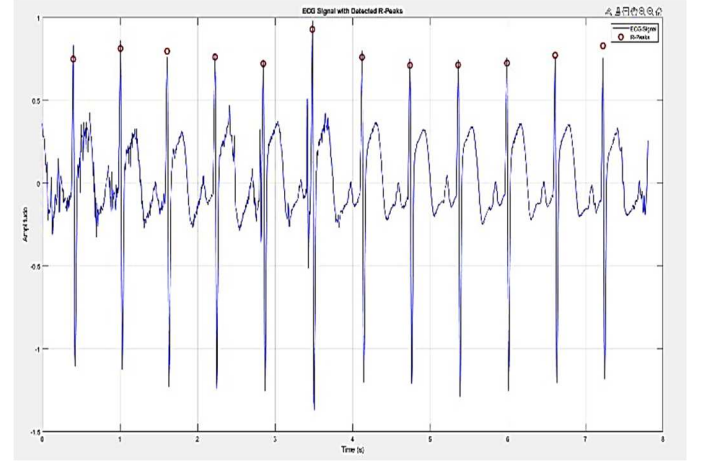


Fig. 3. Representation of R-peaks in ECG signals detected by Pan & Tompkins method

TABLE II. LIST OF 6 TIME AND 1 FREQUENCY DOMAIN FEATURES FOR ECG AND PPG

Feature No	Feature Name	Description
1	MEAN	Mean of R-R SERIES
2	VARIANCE	VARIANCE of R-R series
3	SDNN	SDNN of R-R series
4	MEDIAN	MEDIAN of R-R series
5	NN50	NN50 of R-R series
6	RMSSD	RMSSD of R-R series
7	HRV	Ratio of LF and HF

C. HRV feature extraction and Segmentation:

Heart rate variability (HRV) refers to the fluctuations in the intervals between successive heartbeats. On an electrocardiogram (ECG), the R wave represents the peak of each heartbeat, and the period between two successive R wave peaks is known as the R-R interval. Before analyzing HRV, the ECG must be cleaned up by removing any beats that don't come from the heart's normal pacemaker, called the sinus node. The time between normal beats is then called the N-N (normal-normal) interval. HRV measures how much the N-N intervals vary. The different waves of ECG signal is explained in Fig. 4.

The signal is divided into segments to detect the waves, segments, and intervals. Their behavioral and structural characteristics may be used to correlate them with established patterns. To properly diagnose different cardiac problems, it is crucial to correctly identify distinct parts (segments) of the ECG waveform. Heart rate variation (HRV)[14], which evaluates the regulation of the autonomic nervous system, can be used as a monitoring tool for conditions including cardiovascular diseases, neuropathies, and various phases of sleep.

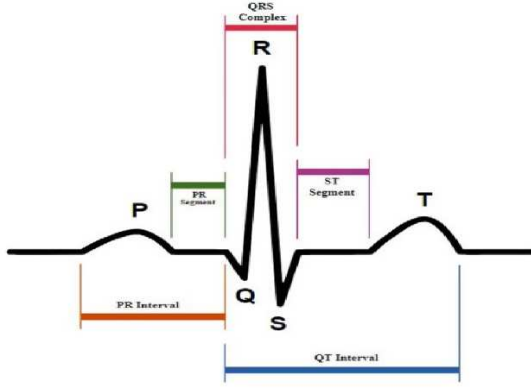


Fig. 4. PQRS, T waves, PR interval, and QT interval on ECG signals [20]

Data from photoplethysmography (PPG) and electrocardiograms (ECG) may be utilized to extract it. According to certain theories, heart rate variation (HRV) is correlated with MW variations and tends to decrease as MW increases[15]. Several time domain characteristics were retrieved, including the mean of R-R series, the median of R-R series, the root mean square of R-R series, the standard deviation of R-R series, and the proportion of R-R intervals that differ from 50ms. The frequency spectrum was analyzed using Welch's periodogram method alongside the Fast Fourier Transform (FFT) to extract valuable frequency domain information. This approach enabled the retrieval of essential characteristics from the spectrum, facilitating a comprehensive understanding of the underlying signal properties. The frequency spectrum was divided into very low frequency (VLF), low frequency (LF), and high frequency (HF) bands to analyze the HF and LF spectra more precisely [14]. In heart rate variability (HRV) analysis, the ratio of the power spectral density (PSD) between the HF and LF bands is commonly used as a key indicator.

$$\text{SDNN} \quad \text{sqrt} \left\{ \frac{\sum_{i=1}^N (RR_i - mRR)^2}{N-1} \right\} \dots\dots\dots (1)$$

$$\text{RMSSD} \quad \text{sqrt} \{ \text{mean}((RR_i + 1 - RR_i)^2) \} \dots\dots\dots (2)$$

$$\text{HRV} \quad \frac{1}{N-1} \sum_{i=1}^N (RR_i) \dots\dots\dots (3)$$

$$\text{Variance} \quad S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} \dots\dots\dots (4)$$

D. Classification:

After extracting features from both ECG and PPG signals, we proceeded with the classification task. Our dataset includes ECG and PPG signals collected from 22 individuals, sampled at 256 Hz. We segmented the signals into 8-second intervals, with each segment containing approximately 10 quasi-cycles for the ECG signals and 12 quasi-cycles for the PPG signals. In total, 14 features were collected (seven from each signal) detailed in Table II. The goal of this study is to classify levels of mental stress among participants, guided by NASA-TLX data. NASA-TLX is a widely validated tool for assessing mental workload through various attributes, offering high accuracy and reliability. For this purpose, we

implemented classification models with three and four classes, segmenting the NASA-TLX weighted values accordingly. In the 3-class model, segments were labeled as LOW, MEDIUM, and HIGH. In the 4-class model, segments were labeled A, B, C, and D, with A representing the highest level of mental workload. We employed several machine learning algorithms to complete the classification, including Random Forest, Decision Tree, SVM, KNN, and Gaussian Naive Bayes (GNB) [16]. We utilized Google Collab to train our machine learning algorithms, creating a dataset with 264 data points based on the features developed for training. The dataset was divided into an 80:20 train-test split, and cross-validation was performed using the k-fold method to enhance model reliability and mitigate overfitting. To evaluate cross-validation performance, we calculated precision, recall, and F1 score, with results illustrated in Figure 5. The results clearly indicate that the random forest algorithm outperformed the other models, with the decision tree ranking second. To mitigate overfitting, cross-validation was performed multiple times with varied data splits. For a more comprehensive performance analysis, we employed several metrics, including the confusion matrix and the area under the curve (AUC). The confusion matrix provides a visual and quantitative summary of classification effectiveness, displaying predicted versus actual values in a two-dimensional format. From this matrix, we obtained true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) for the trained model, which allowed us to calculate precision, recall, accuracy, and AUC.

$$\text{Precision} = \frac{TP}{TP+FP} \dots\dots\dots (4)$$

$$\text{Recall} = \frac{TP}{TP+FN} \dots\dots\dots (5)$$

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN} \dots\dots\dots (6)$$

IV. RESULTS AND ANALYSIS

To evaluate our model, we tested it on separate samples after training on the training set. The entire dataset was divided into two ratios: 80:20 and 70:30 for the training and testing sets. The training set was further split into a validation set using a 70:30 ratio. After conducting the validation tests, we assessed the prediction accuracy of various models.

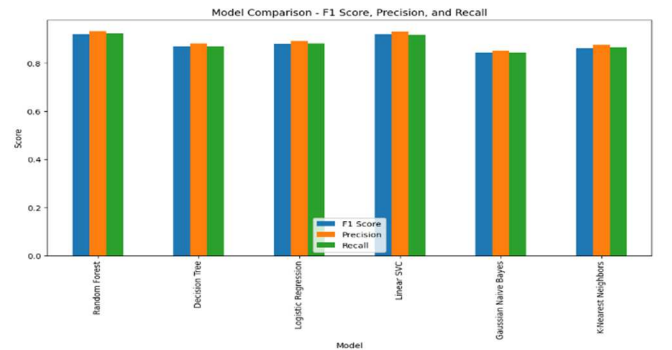


Fig. 5. F1 Score, Precision and Recall for three class classification

Based on the F1 score, the top-performing models for this dataset were the random forest, decision tree, and KNN. To mitigate overfitting, we adjusted the hyperparameters of all classifier models, with particular emphasis on the random forest and decision tree, as they are more susceptible to overfitting than others.

We can see the confusion matrix of Random-Forest classifier for four-class classification in Fig. 6. The graph that displays the effectiveness of a classification model's overall categorization levels is known as ROC. AUC quantifies the extent to which two classes may be distinguished from one other. It quantifies the model's ability to differentiate between different classes. A model with a higher AUC value indicates a stronger ability to accurately predict 0 classes as 0 and 1 classes as 1. By this analogies, it can be understood that a higher Area Under the Curve (AUC) indicates a greater ability of the model to differentiate. We can see the AUC curve of random forest for four-class classification in figure 7.

TABLE III. OVERVIEW OF ACCURACIES FOR PROPOSED MODELS

Classifier	Three Class (%)		Four Class (%)	
	Cross validation set	Test set	Cross validation set	Test set
Random-Forest	93	96	96	95
Decision-Tree	86	89	72	84
Logistic Regression	88	87	88	88
Linear SVC	91	93	89	90
Gaussian NB	80	83	79	82
KNN	83	84	82	83

Table III displays the testing accuracies for the various models we trained. The Random Forest algorithm achieved the highest accuracy in both three-class and four-class classifications. The table provides details on both validations set accuracy and test set accuracy, illustrating the models' performance across different data subsets. The implementation of cross-validation was instrumental in preventing overfitting and minimizing redundancy in the dataset, particularly given the limited number of data points. This approach ensures that the model generalizes well to unseen data, thereby enhancing the robustness and reliability of our classification results.

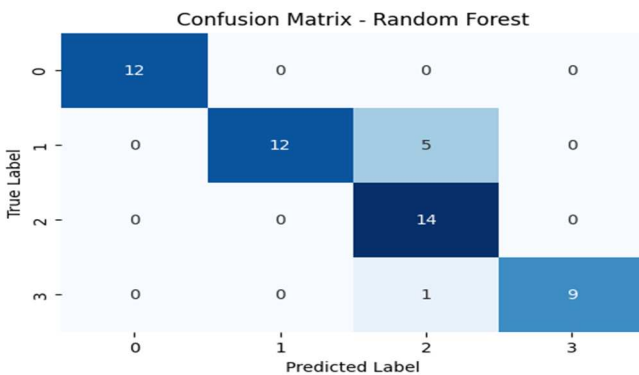


Fig. 6. Confusion matrix of Random-Forest Classifier for four-class

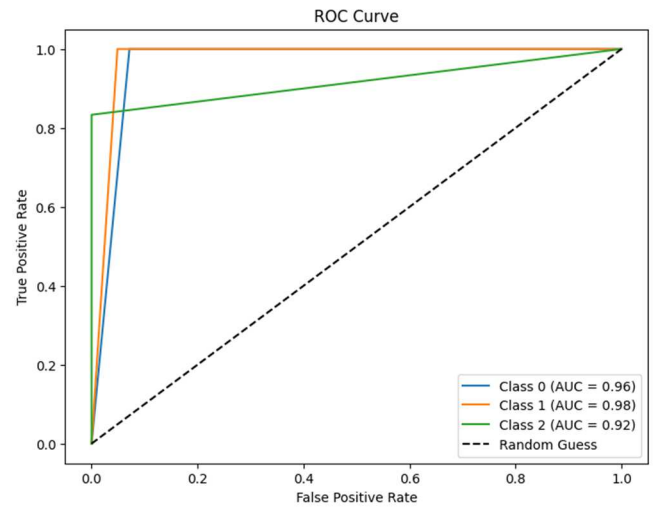


Fig. 7. AUC curve of random forest for four-class classifiers

V. CONCLUSION

Nowadays, mental stress is widespread across people of all age groups due to a continuously demanding and competitive lifestyle. As stress can damage a person's emotions and physical health, early diagnosis is crucial for further action. In recent studies, physiological data signals like ECG, PPG, and GSR have been used to quantify stress. Some studies used one physiological signal, and others used two or more signals to detect mental stress. All created systems first retrieved features using different approaches and then built different models of machine learning algorithms to quantify mental stress. This study shows that it is possible to detect stress by combining two physiological signals (ECG and PPG) to detect mental stress more accurately. Support Vector Machine (SVM) and Logistic Regression have demonstrated the highest accuracy. Specifically, for 3-class classification, the accuracy surpasses that of 4-class classification. Further improvements in accuracy can be achieved by preprocessing the data with outlier analysis for ECG and PPG signals. This can be compared to the recent study, 'Robust PPG-Based Mental Workload Assessment System Using Wearable Devices' by Win-Ken Beh, Yi-Hsuan Wu, and An-Yeu Wu, which employs PPG signals with outlier removal, achieving an accuracy of $(74.2 \pm 10)\%$ [19]. Using ECG signals alone, their system achieves an accuracy of $(75.3 \pm 8.9)\%$. In contrast, our proposed model integrates both ECG and PPG signal features, demonstrating superior accuracy in both three-class and four-class classification, with the best results achieved using the random forest classifier.

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